

AI531C – T07: The Measurement Studio

Objective

The task is to convert a binary image into measurable objects and use area, centroid, aspect ratio, extent, and morphological opening to study which objects survive as the structuring-element (SE) size increases.

1. Binary mask and connected components

The program reads coins.png as a grayscale image and creates a binary mask using Otsu thresholding. Connected components are then detected. Each component represents a candidate object; the background (label 0) is ignored. Very small components are removed as noise.

2. Object measurements

For every remaining object, OpenCV provides its area, bounding-box position, width, height, and centroid. The centroid is the geometric center of the object based on its pixels.

Aspect ratio = $\max(\text{width, height}) / \min(\text{width, height})$. Values close to 1 indicate a roughly compact object, while a large value indicates an elongated object.

Extent = object area / bounding-box area. A compact object fills more of its bounding box and therefore has a larger extent. The T07 sheet reports coins at about 0.77 and the impostor band at about 0.42, making extent useful for separating the two.

3. Structuring-element sweep

The program applies morphological opening with elliptical structuring elements of sizes 5, 25, 35, and 45. Opening consists of erosion followed by dilation. As the SE becomes larger, objects that are too small to survive the morphological operation disappear. The supplied T07 sheet gives the expected sweep as 24, 23, 2, and 2 survivors respectively.

4. Why the gate is important

The sharp change between SE 25 and 35 is the 'size gate'. The sheet connects this transition to the median coin area of 1492 pixels. Approximating a coin as a circle gives:

$$d \approx \sqrt{(4A/\pi)} = \sqrt{(4 \times 1492 / \pi)} \approx 44 \text{ pixels}$$

5. Area filtering vs. morphological opening

Area filtering removes connected components directly according to their measured area. Morphological opening instead changes the binary shape using a structuring element, so survival depends on object size and shape relative to the SE. Therefore, the two methods can remove small objects for different reasons.

Expected result / conclusion

The program demonstrates the complete measurement pipeline: image → binary mask → connected components → measurements → decisions. The aspect and extent values help distinguish the elongated impostor from coins, while the SE sweep demonstrates that morphological opening can produce a sharp size threshold rather than a gradual reduction in the number of surviving objects.